

# Akshay Shivashankar Bharadwaj

Los Angeles, CA | ab79035@usc.edu/akshaysbharadwaj2k@gmail.com | +1 213 301 6674  
ab79035.github.io/Portfolio | github.com/akshaysb2k | linkedin.com/in/akshay-s-bharadwaj2k

## Education

- University of Southern California**, MS in Computer Science (GPA 3.86) Jan 2025 – Dec 2026
- **Courses:** Analysis and Design of Algorithms, Web Technologies, Operating Systems, Cloud and Distributed Systems
- BMS College of Engineering**, BE in Computer Science (GPA 4.0) Aug 2018 – Jun 2022
- **Courses:** Data Structures and Algorithms, Web and Database design, Artificial Intelligence, Machine Learning

## Experience

- IT Engineer**, Qualcomm - Bengaluru, India June 2022 – Dec 2024
- Engineered a **ReactJS-Flask** internal tool integrating **ServiceNow APIs** to automate end-to-end RFC (request for change) workflows, eliminating **100+ engineering hours of manual effort per month**.
  - Reduced deployment time by **50%** by building containerized microservices infrastructure using **Docker, Kubernetes (EKS), CloudFormation, and Jenkins CI/CD pipelines**, eliminating environment-parity issues across S3, RDS, Route53, VPC, and Lambda.
  - Drove **cross-functional collaboration** across teams to architect secure infrastructure for **Me2U** (Qualcomm's global file transfer platform) using **VPC endpoints, DynamoDB, and S3 lifecycle policies** for thousands of internal and external users without public internet routing.
  - Designed an **AWS CloudFormation automation** framework to onboard **70+ web applications** on EC2 with auto-scaling with MFA and 24/7 CloudWatch/DataDog alerting to enforce SLAs and leveraged **GenAI tools** to drive cloud cost optimization by over **60%**.
- Interim Intern**, Qualcomm - Bengaluru, India Feb 2022 – Jun 2022
- Built a **self-service AWS deployment infrastructure using CloudFormation and Lambda** that enabled teams to deploy Java applications with minimal manual input, cutting new environment setup from days to **under an hour**.

## Projects

- Operating Systems Kernel Development: Weenix (Unix-Like OS Kernel)** Aug 2025 – Dec 2025
- Designed and implemented process and thread management in C, including scheduling, context switching, and synchronization.
  - Achieved full system call support by building a **virtual file system (VFS)** within a Linux-inspired OS kernel architecture.
  - Ensured robust memory isolation by implementing **virtual memory mapping** and **page fault handling** for concurrent user processes.
  - Supported concurrent process creation by developing `fork()` based child process creation in the Weenix kernel.
- Raft Consensus Protocol: Distributed Systems (C++20, gRPC)** Jan 2026 – May 2026
- Achieved **99.9% uptime** by implementing **Raft consensus leader election and log replication protocols** in C++20 using gRPC.
  - Guaranteed safety and liveness under node crashes, network partitions, and delayed RPC delivery across simulated failure scenarios.
  - Eliminated race conditions by designing concurrent, **thread-safe state management** for distributed RPC handlers.
  - Achieved **zero data loss** and **sub-500ms leader re-election** validated via fault-injection tests under real-world failure conditions.
- Artsy: Web & Mobile App using Django, PyTorch, Angular.js, Android Studio and PostgreSQL** Jan 2025 – May 2025
- Delivered **sub-200ms API response times** by building an **Angular.js** frontend with a **Django REST Framework** API and GCP deployment.
  - Achieved **87% accuracy** by training a **PyTorch** image classifier on artwork metadata scraped and normalized using **BeautifulSoup**.
  - Built an **Android Studio** mobile client consuming the REST APIs to deliver real-time artwork discovery on mobile.
- Eye-Move: Eye-Tracking Based Typing Application (Computer Vision)** Oct 2021 – Jun 2022
- Enabled hands-free communication for motor-neuron disability patients by building a virtual keyboard using **OpenCV and DLib**.
  - Achieved **87.5% typing accuracy** by developing precise blink detection algorithms calibrated for varied lighting conditions.
  - Reduced communication barriers for disabled users by optimizing detection latency and error thresholds in real-time.
- Moonquake Alert System (MAS), Space@yourService - EPFL (Swiss Federal Institute of Technology)** Jul 2020 – May 2021
- Achieved **92% detection accuracy** by building a **Python-based** moonquake detection system using signal processing and anomaly detection.
  - Reduced false positives by building a **multi-stage anomaly detection pipeline** using **bandpass filtering** on raw lunar rover seismic data.
  - Guaranteed **astronaut safety** by containerizing the detection system on **Docker** in coordination with EPFL for the Asclepius experiment.
  - Achieved **sub-5 second alert dispatch** by decoupling the detection engine from the notification layer via a **config-driven architecture**.

## Technologies

**Languages:** Python, Go, C++, C, C#, Java, SQL, JavaScript, TypeScript, PHP, HTML, CSS, Unix Shell Scripting, Linux Kernel Programming  
**Frameworks & Libraries:** Flask, Django, PyTorch, REST APIs, Node.js, Express, ReactJS, AngularJS, Bootstrap, Tomcat, BeautifulSoup  
**Tools:** AWS, Docker, Kubernetes, Terraform, Jenkins, Ansible, Git, Bash, Datadog, PowerBI, PagerDuty, MongoDB, MySQL, PostgreSQL

## Certifications

AWS Certified Solutions Architect – Associate, Amazon Web Services

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